
Does correction recover the future feature?

Week 6 · an oracle diagnostic splits feature recovery from output recovery, and claim C1 weakens.

Cached Future-Feature Prediction

Weekly research update · synthetic demo · version 0.1

Where we are, and what changed this week

ESTABLISHED (UNCHANGED)

- **P1** Cached hidden features go stale; reusing them as future features drifts.
- **Method** Cache reuse followed by a learned correction toward the future feature.
- **Baseline** Reuse only scores rollout cosine 0.42.

WHERE WE ARE NOW

Method v4 adds a correction objective and an oracle diagnostic. The question this week is whether the correction moves toward the future feature, or whether a different downstream-valid direction is enough.

Last week: v3

What changed in the method: v3 to v4

v3 → **v4** v4 adds a correction objective and an oracle diagnostic, and scales down the correction.

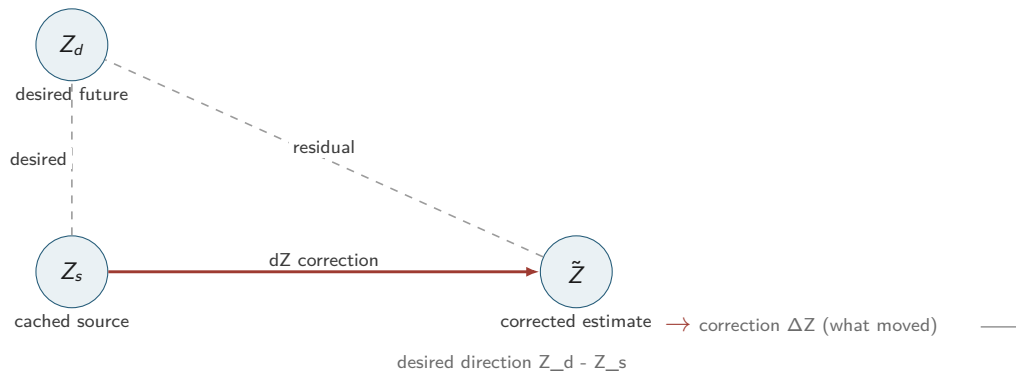
CHANGES THIS WEEK

- Added a correction objective
Penalize deviation from the future feature so the correction is supervised directly.
- Added an oracle diagnostic
Separate feature recovery from output recovery to test C1 and C2.
- Scaled down the correction magnitude
Reduce drift when the correction is applied over long rollouts.

UNCHANGED

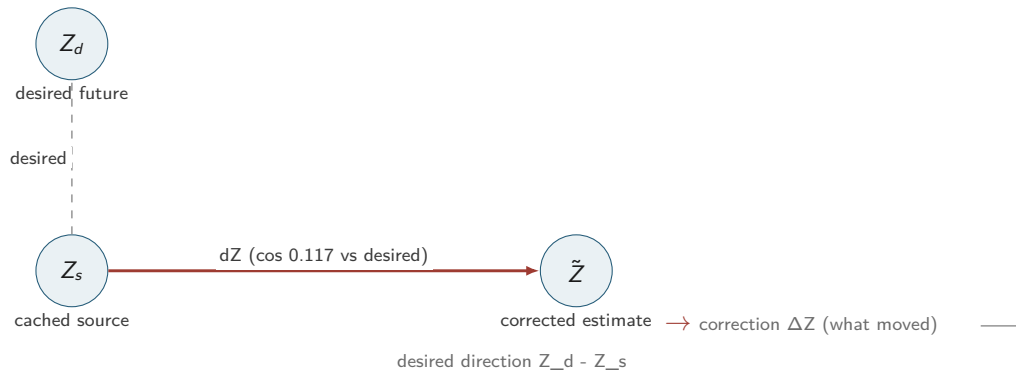
- Cache layout and update schedule
- Backbone and dataset split
- Rollout evaluation protocol

Two stories for what the correction does



C1 says dZ follows the desired direction; C2 says dZ can be orthogonal and still be useful.

Correction is nearly orthogonal to the desired direction



Same objects and positions as the previous slide; the measured dZ is ~ 90 degrees from desired.

Diagnostic D1: does the correction aim at the future feature?

D1 Does the correction direction point toward the future feature?

tests C1

MEASUREMENT

$$\cos(\text{deltaZ}, Z_d - Z_s) = 0.1168$$

OBSERVATION

The correction direction is nearly orthogonal to the direction from the cached feature to the future feature.

INTERPRETATION

Output recovery may not require moving toward the exact future hidden state.

CAN CONCLUDE

C1 as stated is not supported; the correction is not simply moving toward Z_d .

CANNOT CONCLUDE

That the correction is uninformative, or that the method fails.

ALTERNATIVE TARGETED

The correction may exploit a downstream-valid direction that is not the future-feature direction.

Comparison: competitor, last week, this week

Method	Rollout cosine	Downstream acc.	Feature recovery R2
Cache reuse only	0.42	0.74	0.28
Ours (v3, last week)	0.51	0.83	0.30
Ours (v4, this week)	0.58 (+0.07)	0.91 (+0.08)	0.34 (+0.04)

higher is better v4 improves both downstream metrics over v3. Feature recovery stays low; that is the evidence behind the weakened C1 and the emerging C2, not a hidden weakness.

What this week does to our claims

C1 plausible → weakened

The learned correction approximates the true future feature.

D1: $\cos 0.117$

C2 absent → emerging

An alternative downstream-valid pre-image may be sufficient; output recovery does not require feature recovery.

D2: high output, low feature recovery Neither claim is refuted or confirmed; both remain under test.

Diagnostic D2: does output recovery require feature recovery?

D2 Does output recovery require feature recovery?

tests C2

MEASUREMENT

linear probe R^2 for recovering Z_d from the corrected feature = 0.34; downstream accuracy = 0.91

OBSERVATION

Downstream accuracy is high while feature recovery remains low.

INTERPRETATION

A downstream-valid pre-image that is not the true future feature may be sufficient.

CAN CONCLUDE

Output recovery can occur without strong feature recovery in this setup.

CANNOT CONCLUDE

That no valid feature recovery exists, or that the probe is a complete measurement.

ALTERNATIVE TARGETED

The probe may fail to decode information that is present in the feature.

Limitations and open questions

CURRENT LIMITATIONS

- Alternative valid pre-images have not been directly characterized.
- The oracle diagnostic uses the future feature as an upper bound and is unavailable at inference.
- Feature recovery is measured with a single linear probe.

OPEN QUESTIONS

- Can we characterize downstream-valid pre-images without the oracle?
- Does the correction generalize to horizons beyond the training horizon?
- Would the same orthogonality appear for other cached-feature tasks?